

March 2021

ADVANCED STRENGTH OF MATERIALS

Time Allowed: 2 Hours

Full Marks: 35

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any ten): 1x10

State whether the following statements are True or False:

- i) The strength of the beam mainly depends on the bending moment.
- ii) Strain energy is set up in elastic material.
- iii) Moment of resistance of beam is independent of its section modulus.
- iv) If the section modulus of a beam is increased, the bending stress of the beam will be decrease.
- v) Proof resilience is the maximum strain energy upto elastic limit.
- vi) Spring index is the ratio of mean coil diameter to the diameter of spring.
- vii) For a circular section shear stress distribution is parabola.
- viii) The unit of section modulus and area are same.

Fill in the blanks:

- ix) The bending equation is _____.
- x) The bending stress on a beam section is maximum at _____.
- xi) For a T-section beam, maximum shear stress occurs at _____.
- xii) The angle between the principle planes is _____.
- xiii) The radius of Mohr's stress circle indicates maximum _____ stress.
- xiv) Strain energy is set up in _____ material.
- xv) Proof resilience per unit volume is called _____.
- xvi) The unit of modulus of elasticity is same as of _____.

2. An I-section beam of flanges $20 \text{ cm} \times 2 \text{ cm}$ and web $30 \text{ cm} \times 1 \text{ cm}$ is acted on by a shear force of 150 kN determine: i).the maximum and minimum shearing stress in the web. ii).the maximum shear stress in the flange. 5
3. The principal stress at a point in the section of a boiler shell are 80 N/mm^2 and 40 N/mm^2 . Find the normal, tangential and resultant stress across a plane through the point inclined at 50° to the plane carrying 80 N/mm^2 stress. 5
4. Deduce torsion equation i.e. $\frac{T}{I_p} = \frac{\sigma_s}{R} = \frac{G\theta}{L}$ for solid shaft. 5
5. A solid shaft of 80 mm diameter is to be replaced by a hollow shaft of external diameter of 100 mm. Determine the internal diameter of hollow shaft if the same power is to be transmitted by the both shafts at the same angular velocity and shear stress. 5
6. A closed coiled helical spring having the mean coil diameter is 12 times than the wire diameter. The spring is to absorb 30000 N cm of energy with an extension of 15 cm. The maximum shear stress set up in the spring is not to exceed 10000 N/cm^2 . Determine – (a) mean coil diameter, (b) number of turns. 5
7. What is simple bending? What are the assumptions made in the simple bending? 2+3

8. A steel plate 12 mm thick is bent around a drum. If the maximum stress induced in the plate due to bending be 400 N/mm^2 , find the radius of the drum. Take $E = 2 \times 10^5 \text{ N/mm}^2$. 5
9. Strain energy of two elastic bars subjected to gradual loading having cross-sectional area are 400 mm^2 and 800 mm^2 are same. Establish a relation between stresses of two bars. 5
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